

Chloe Barker

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EDUCATION

Southern Methodist University | Dallas, TX Jan 2025 - Present
M.S. in Data Science | GPA: 3.97 (Expected Aug 2026)

The University of Texas at Austin | Austin, TX Aug 2020 - May 2024
B.S.A. in Biology
Minor: Health Communication; Pre-Health Professions Certificate | GPA: 3.77

TECHNICAL SKILLS

Programming & Tools: Python (pandas, NumPy, scikit-learn, matplotlib, seaborn), R (tidyverse, ggplot2, caret), SQL (MySQL, SQL Server; joins, aggregations, optimization), SAS, Git/GitHub, Jupyter, VS Code

Machine Learning & Modeling: Regression (Linear, LASSO), Classification (Logistic Regression, Random Forest, KNN, LDA/QDA), Time Series, NLP (TF-IDF, clustering, sentiment analysis), Feature Engineering, Cross-Validation, Model Selection (AIC/BIC), Model Evaluation (ROC/AUC, confusion matrices)

Statistics: Hypothesis Testing, ANOVA, A/B Testing, Experimental Design, Multivariate Analysis, Reproducible Research

Visualization & BI: Tableau, R Shiny, Data Storytelling, Executive Presentations

Data & Productivity Tools: Excel (pivot tables, data analysis), Google Sheets

WORK EXPERIENCE

Applied Analytics Group, LLC | Remote Jan 2026 - Present
Data Scientist (Part-Time)

- Contribute to early-stage data science and analytics projects using Python, R, and SQL.
- Structure and document GitHub repositories to improve clarity, usability, and presentation of technical work.

Outlier AI | Remote Aug 2024 - Jan 2026
AI Math Evaluator

- Evaluated AI mathematical reasoning across algebra, calculus, and statistics to improve model accuracy.
- Designed structured test cases and error analyses to enhance reliability of reasoning-based AI systems.

RUSH University, Department of Microbial Pathogens and Immunity | Chicago, IL Jun 2023 - Aug 2023
Research Assistant

- Analyzed microbiome and immune-response datasets using Excel and R to study alcohol and circadian effects.
- Cleaned, visualized, and documented experimental data to support reproducible biomedical research.

The University of Texas at Austin, Nutrigenomics Biological Sciences Lab | Austin, TX Aug 2021 - Dec 2021
Undergraduate Researcher

- Investigated effects of artificial sweeteners on bacterial diversity using microbiome data and statistical methods.
- Conducted trend analysis and visualizations in R and Excel to communicate findings to research teams.

RUSH University, Department of Microbial Pathogens and Immunity | Chicago, IL Jul 2021 - Aug 2021
Research Assistant

- Analyzed longitudinal datasets on HIV medication impacts using descriptive statistics and data cleaning.
- Built visual summaries and maintained precise documentation supporting ongoing clinical research.

PROJECTS

MS Capstone: Groundwater Lithium & Dementia Prevalence (Python)

- Analyzed 254 Texas counties using TWDB and Medicare data; engineered multi-year exposure features and built OLS models with robust inference methods (HC3, bootstrap, permutation testing).
- Identified a consistent negative association between lithium exposure and dementia prevalence, supported by geospatial analysis and leave-one-out validation.

Texas Groundwater Database & Geospatial Analysis (MySQL, Python)

- Designed a normalized relational database (64 tables, 3NF) and developed complex SQL queries for large-scale environmental data.
- Integrated SQL and Python (GeoPandas, Folium) to map contamination patterns and identify geographic trends.

Employee Attrition Analysis (R)

- Modeled employee turnover using classification techniques (Naive Bayes, KNN), achieving 78% classification accuracy.
- Identified key drivers of attrition and quantified business impact through statistical testing and cost analysis.